

Repair Manuals

Timing chain: removal/installation

Warnings and recommendations

Unless otherwise advised by the manufacturer, the following procedures are recommended:

Always renew the timing chain

If the timing chain is to be re-used, mark the running direction before removal

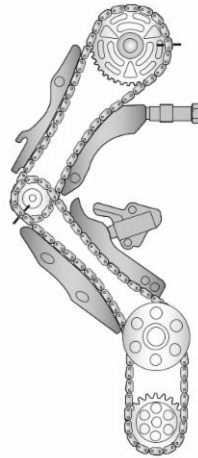
Never turn the engine in the direction opposite to that of normal operation

Note: Avoid bending, torsion or traction

Always use new gaskets

General

Timing chain layout



Removal

Remove the cowl lower grille

Disconnect the battery earth cable

Remove the engine together with the gearbox

Remove the engine upper cover

Remove the resonator

Remove the engine acoustic cover

Remove the turbocharger air inlet pipe

Remove the differential pressure sensor

Disconnect the remaining wiring, cables and hoses

Remove the turbocharger front heat shield

Remove the diesel particulate filter

Remove the turbocharger rear heat shield

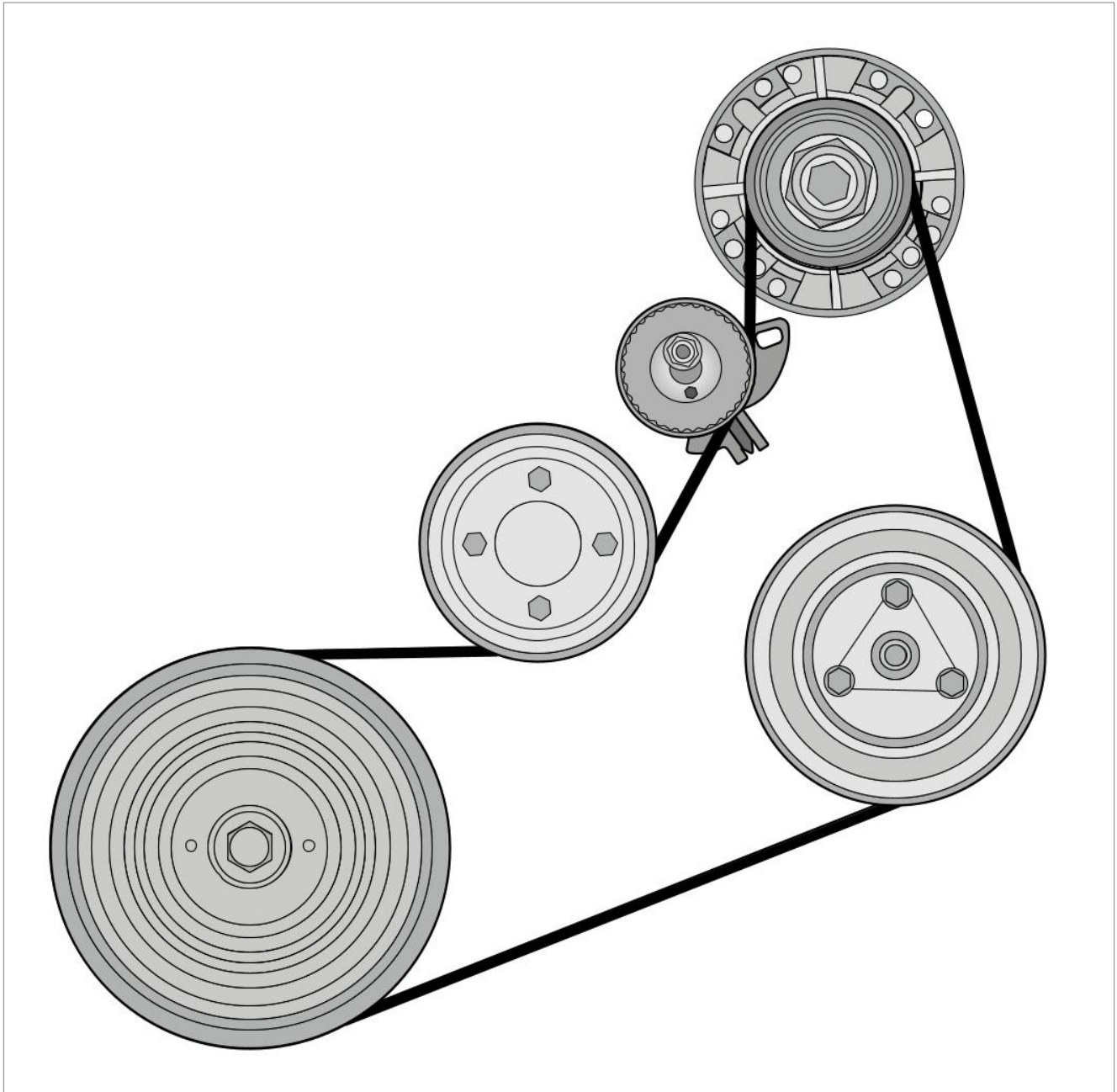
Remove the turbocharger oil supply line

Remove the turbocharger oil return line

Remove the turbocharger

Remove the ancillary drive belt

Drive belt layout

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Remove the alternator

Remove the air-conditioning compressor

Disconnect all the coolant hoses, one by one

Remove the EGR cooler

Remove the engine/gearbox attaching bolts

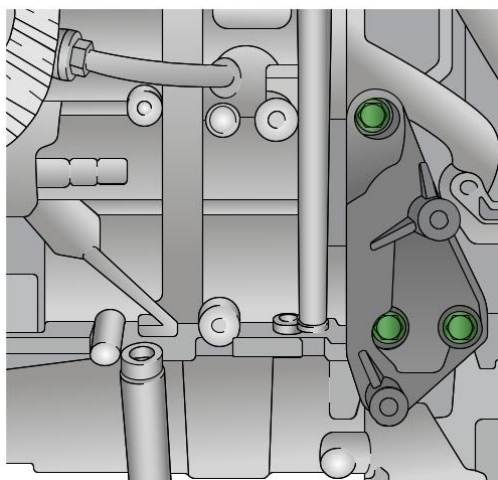
Separate the engine from the gearbox

Fit a lifting eye to the cylinder head ((21.5 Nm)) (2 459 012)

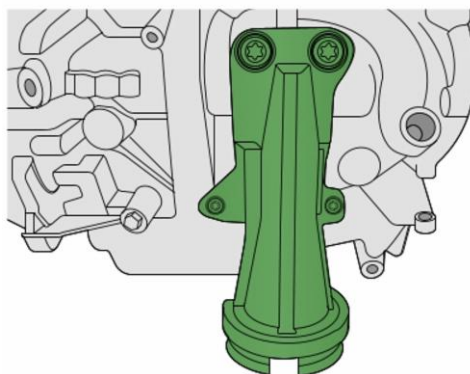
Use a hoist to support the engine (11 0 020, 2 220 718)

Note: Lift the engine

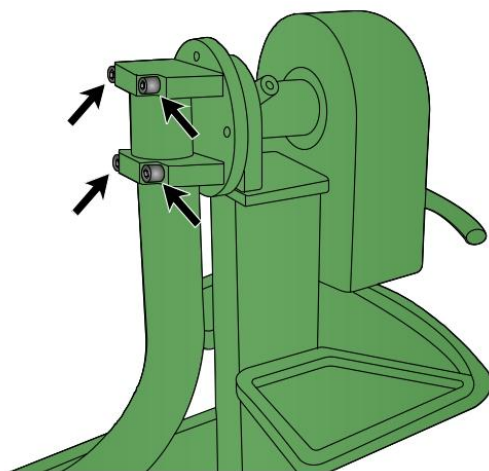
Remove the engine support bracket

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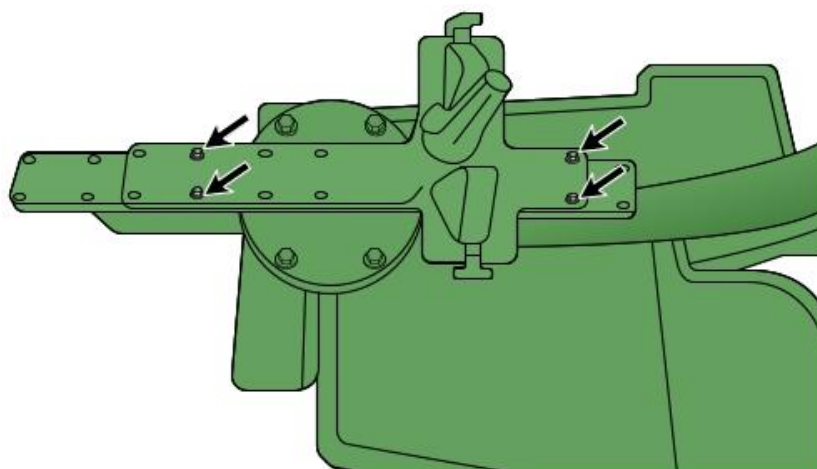
Fit the special tool ((38 Nm)) (2 455 010)



Assemble the special tool as shown. Tighten the special tool bolts ((130 Nm)).

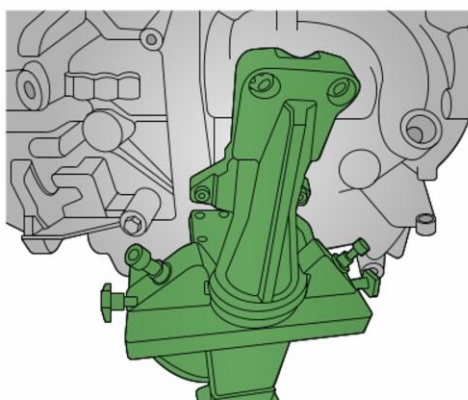


Fit the special tool ((160 Nm)) (2 413 082)

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Lower the engine

Tighten the special tool (2 413 082, 2 455 010)

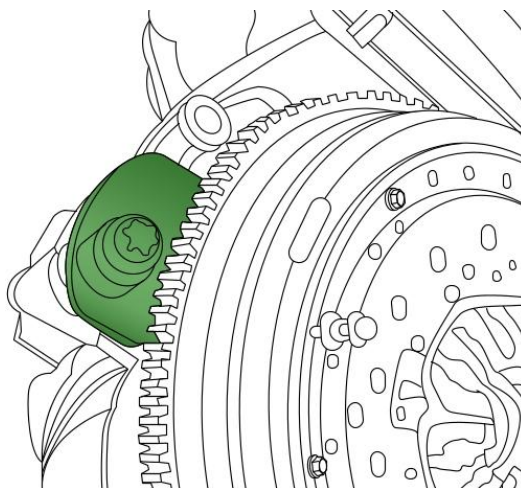


Remove the cylinder head

Remove the crankshaft TDC locking pin (2 288 380)

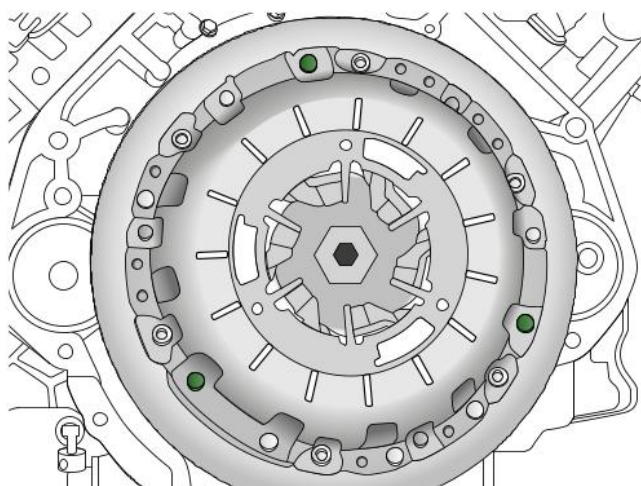
Vehicles with manual transmission:

Fit the locking tool (11 9 265, 11 9 811)

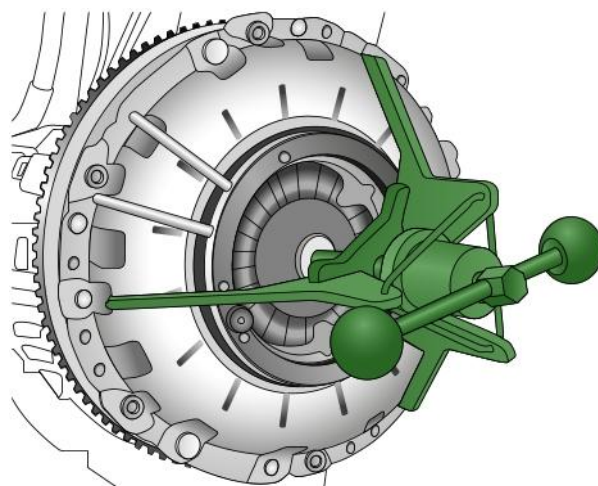


Loosen the three bolts

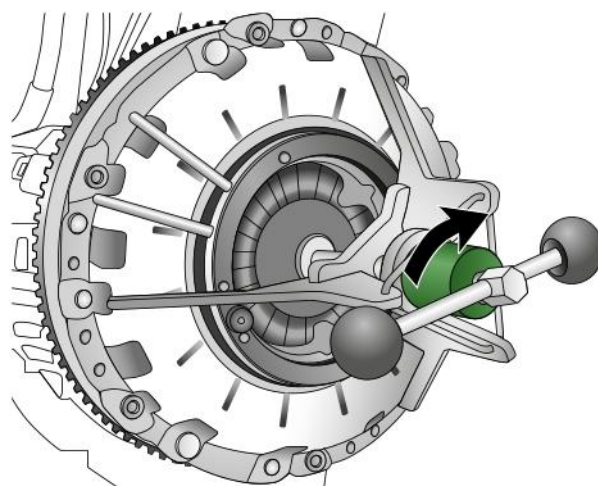
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Fit the special tool (21 2 170)

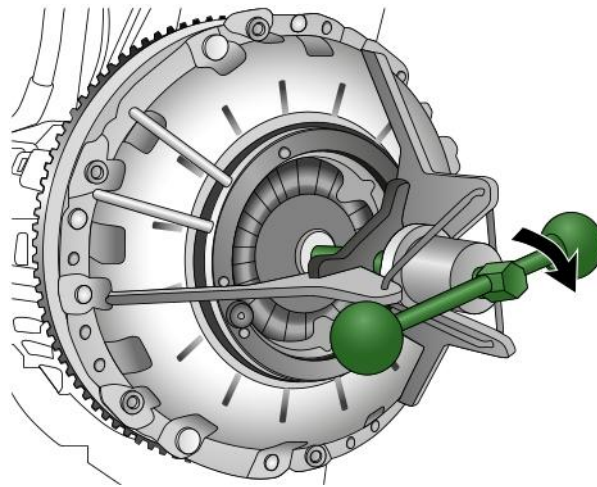


Tighten the screw as far as the stop



Tighten the diaphragm spring to its stop

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Remove the clutch pressure plate bolts

Remove the clutch

Remove the flywheel (11 4 180)

Vehicles with dual-clutch transmission:

Fit the locking tool. With washer: (5 mm).

Remove the drive plate (11 4 180)

All vehicles:

Remove the crankshaft position sensor gearwheel

Remove the dipstick tube

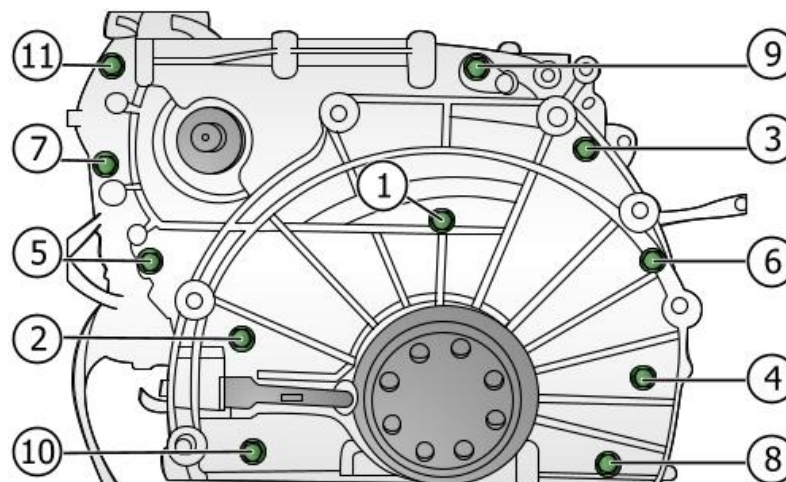
Remove the oil sump protective cover

Remove the oil sump

Disconnect the coolant hoses from the timing cover

Remove the wiring loom bracket

Remove the bolts in the reverse order to that shown ((11 -> 1))



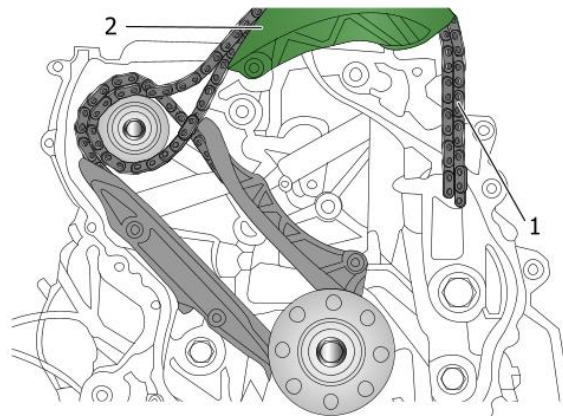
Remove the timing chain cover. Use a screwdriver.

Remove the oil pump chain

Remove the secondary timing chain ((1))

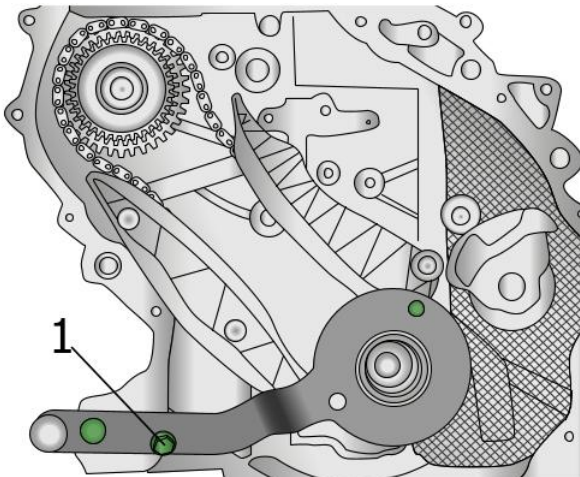
Remove the chain tensioner guide ((2))

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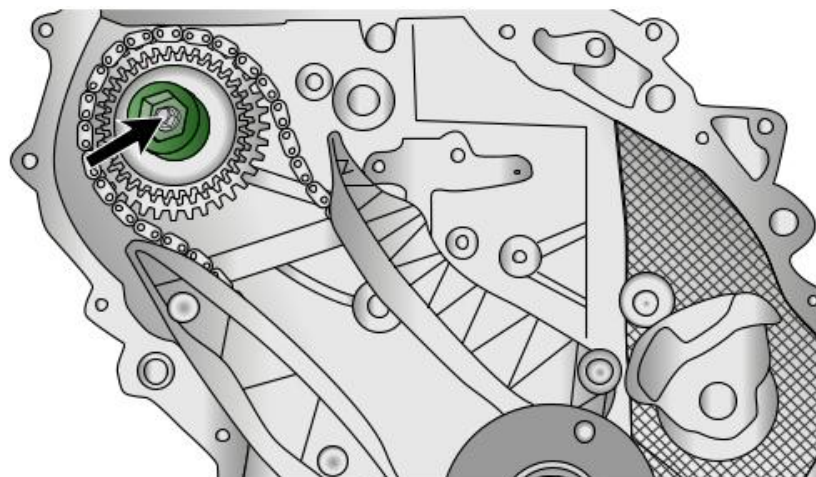
Fit the special tool at the point shown (2 357 904)

Secure the special tool with a bolt ((1))



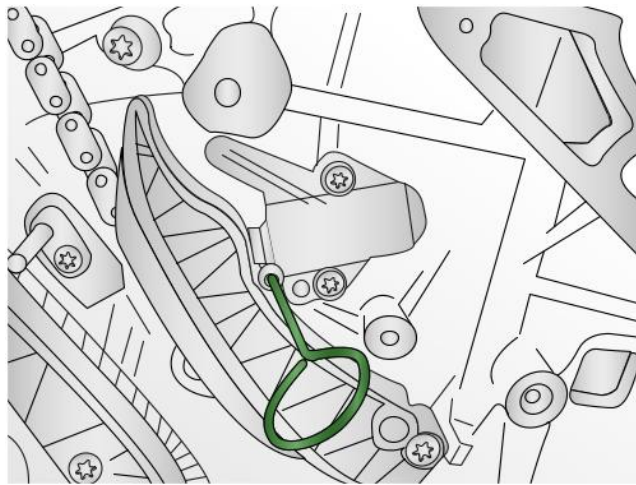
Loosen the high-pressure pump gearwheel bolt

Fit the special tool. Detach the high-pressure pump gearwheel.



Compress the primary chain tensioner

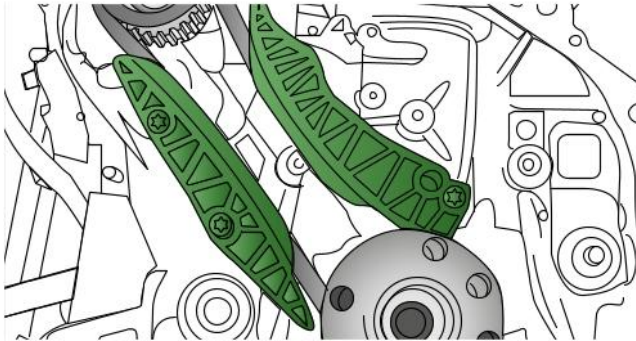
Lock the chain tensioner with a locking pin (11 4 120)

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Remove the primary chain tensioner

Remove the chain guide

Remove the chain tensioner guide



Remove the high-pressure pump gearwheel

Remove the crankshaft locking tool (2 357 904)

Remove the primary timing chain

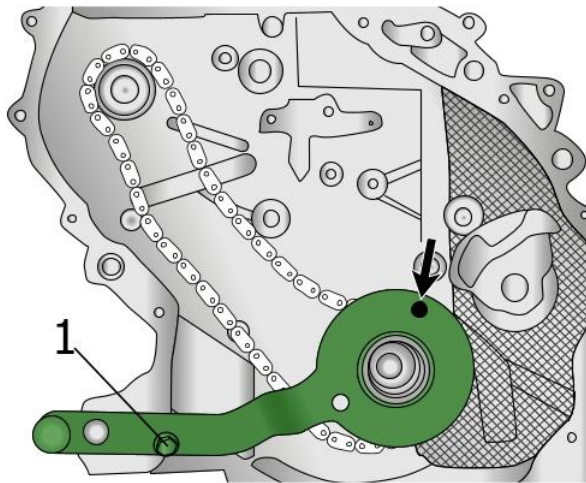
Installation

Fit the primary timing chain

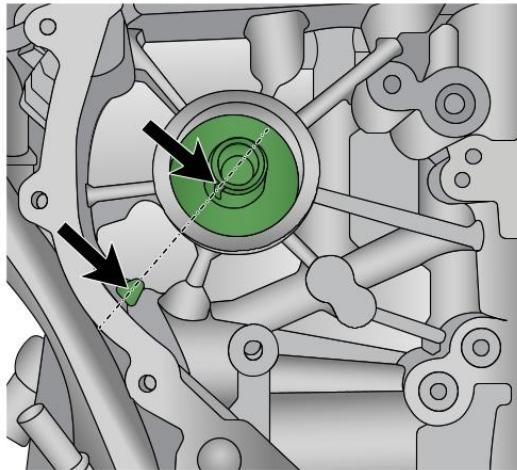
Fit the special tool at the point shown (2 357 904)

Secure the special tool with a bolt ((1))

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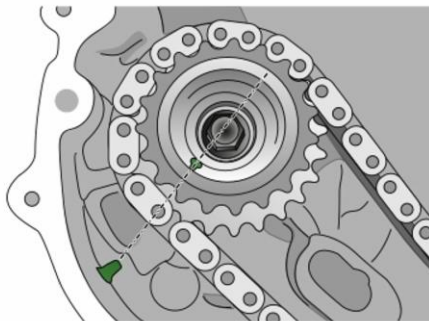


Check the timing marks



Refit the high-pressure pump gearwheel

Note: Do not tighten the bolt



Refit the chain tensioner guide

Refit the chain guide

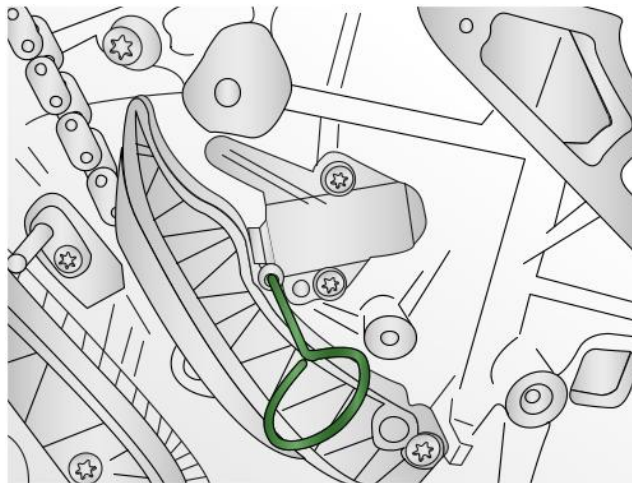
Note: Use new bolts

Refit the primary chain tensioner

Note: Clean off the sealing surfaces

Remove the tensioner locking pin (11 4 120)

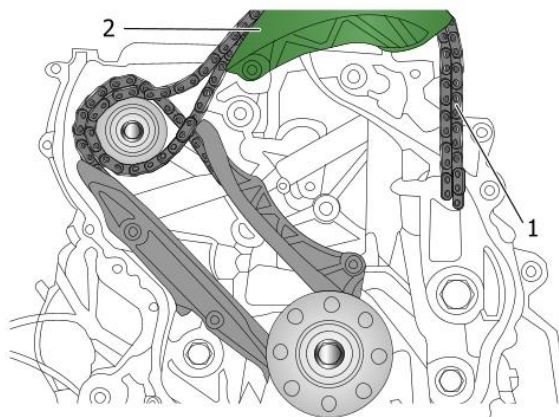
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Tighten the high-pressure pump gearwheel

Fit the secondary timing chain ((1))

Refit the chain tensioner guide ((2))



Fit the oil pump chain onto the crankshaft gearwheel

Refit the oil pump gearwheel

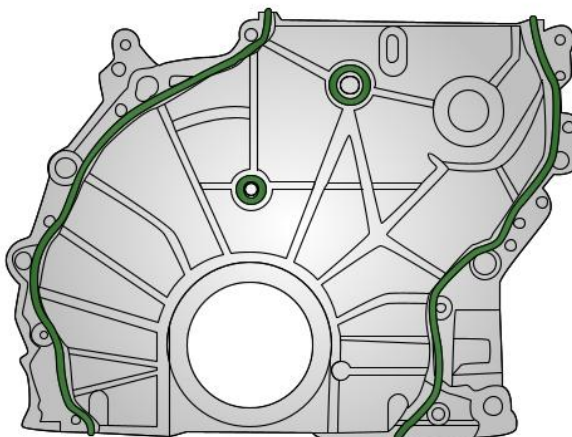
Timing chain cover:

Clean the contact surfaces

Check that the dowel pins are still in position

Apply sealant as indicated ((1.5 - 2.0 mm))

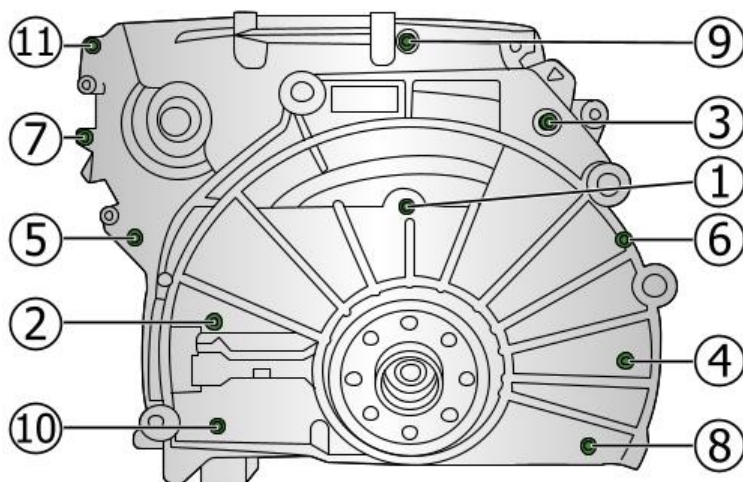
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Note: The cover must be fitted within 10 minutes of applying the sealant

Tighten the timing cover bolts

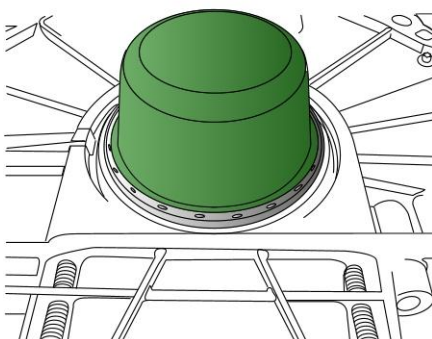
Note: Use new bolts ((1 - 11))



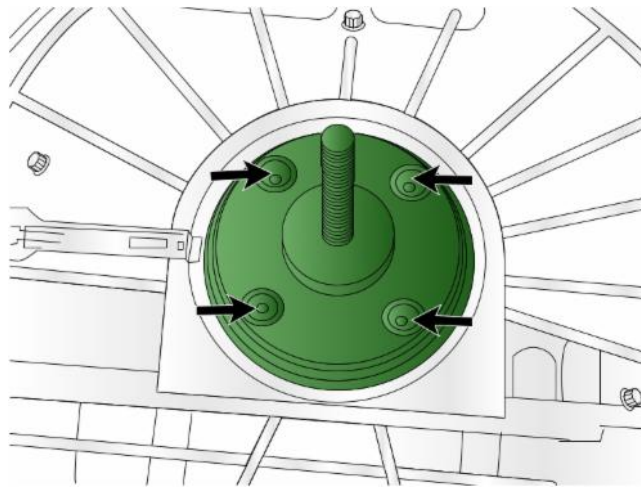
Note: Do not touch the lip of the seal

Fit the crankshaft oil seal

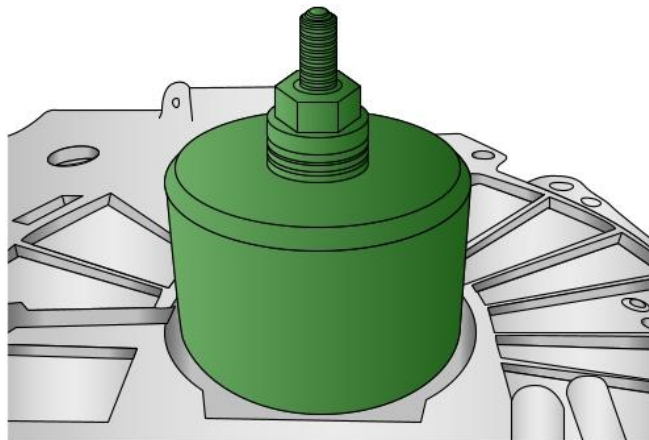
Use the special tool (11 8 815)



Fit the special tools (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)

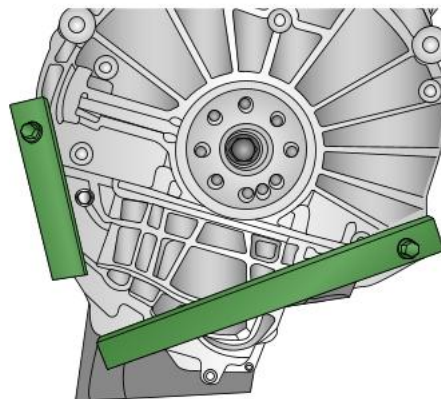
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Tighten the special tool bolt until full surface contact is made



Remove the special tools (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)

Refit the oil sump. Use the special tool.



Further refitting is in the reverse order of removal

Reconnect the battery earth cable

Torque settings

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Timing chain tensioner

Timing chain tensioner (Upper) 70 (Nm)

Timing chain tensioner (Lower) 9 (Nm)

Timing chain tensioner guide

Timing chain tensioner guide 20 (Nm)

Timing chain guide

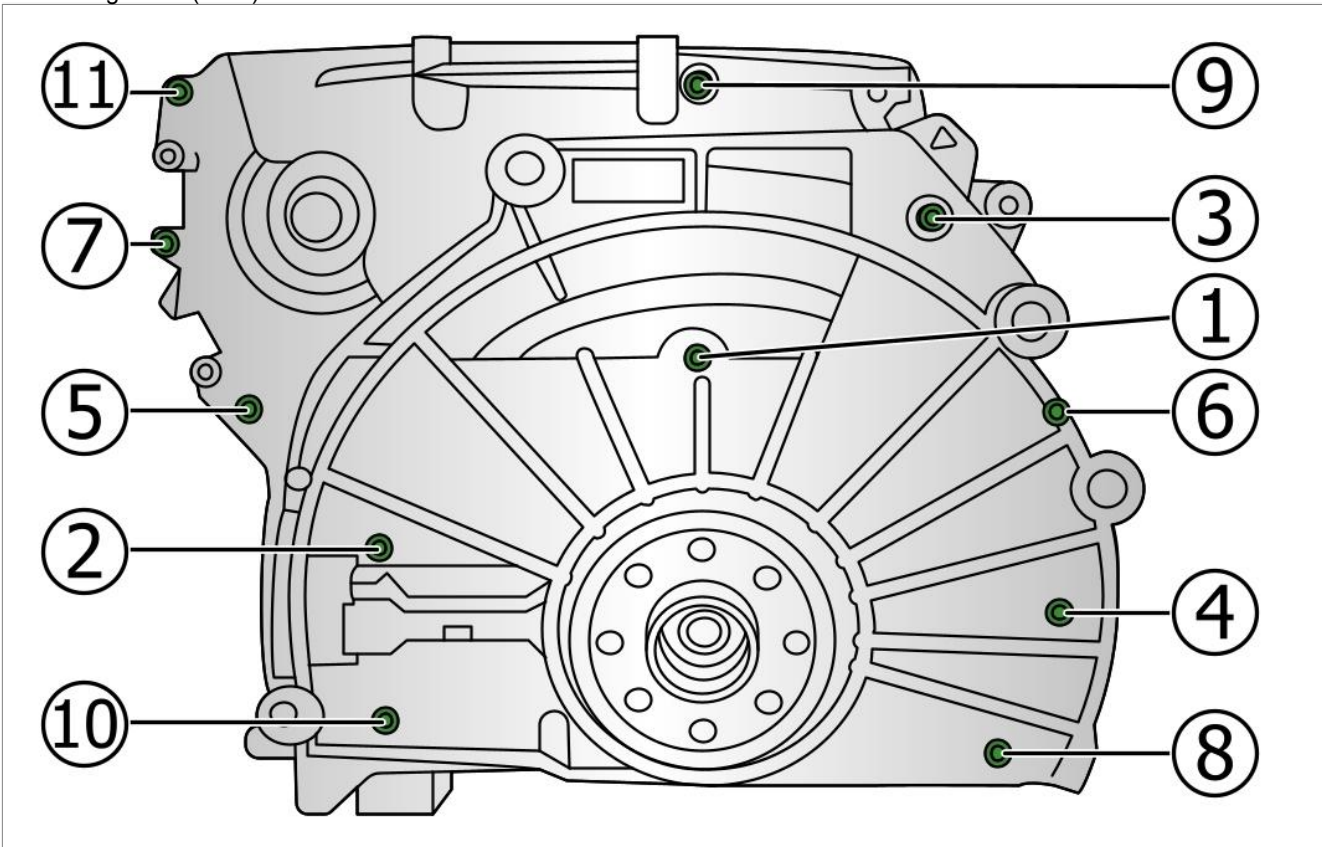
Timing chain guide 20 (Nm)

High-pressure pump gearwheel

High-pressure pump gearwheel 65 (Nm)

Timing cover

Timing cover (Rear)



Renew the bolts

Stage 1 8 (Nm)

Stage 2 90 (°)

Timing cover (Front)

Renew the bolts (Tighten the bolts in a criss-cross pattern)

Stage 1 4 (Nm)

Stage 2 60 (°)

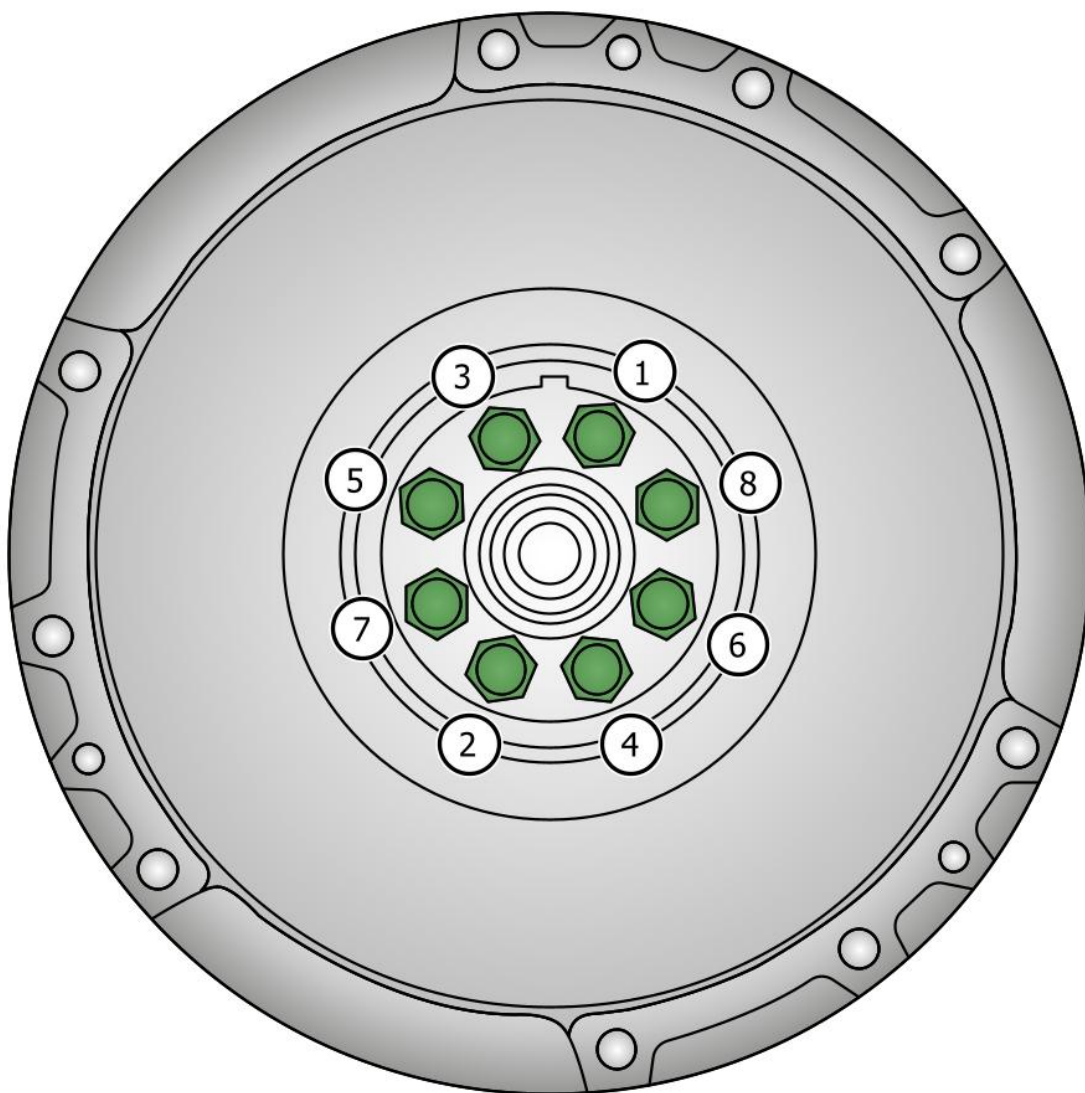
Oil sump

Sump 24 (Nm)

Flywheel

Flywheel

Renew the bolts

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Stage 1 45 (Nm)

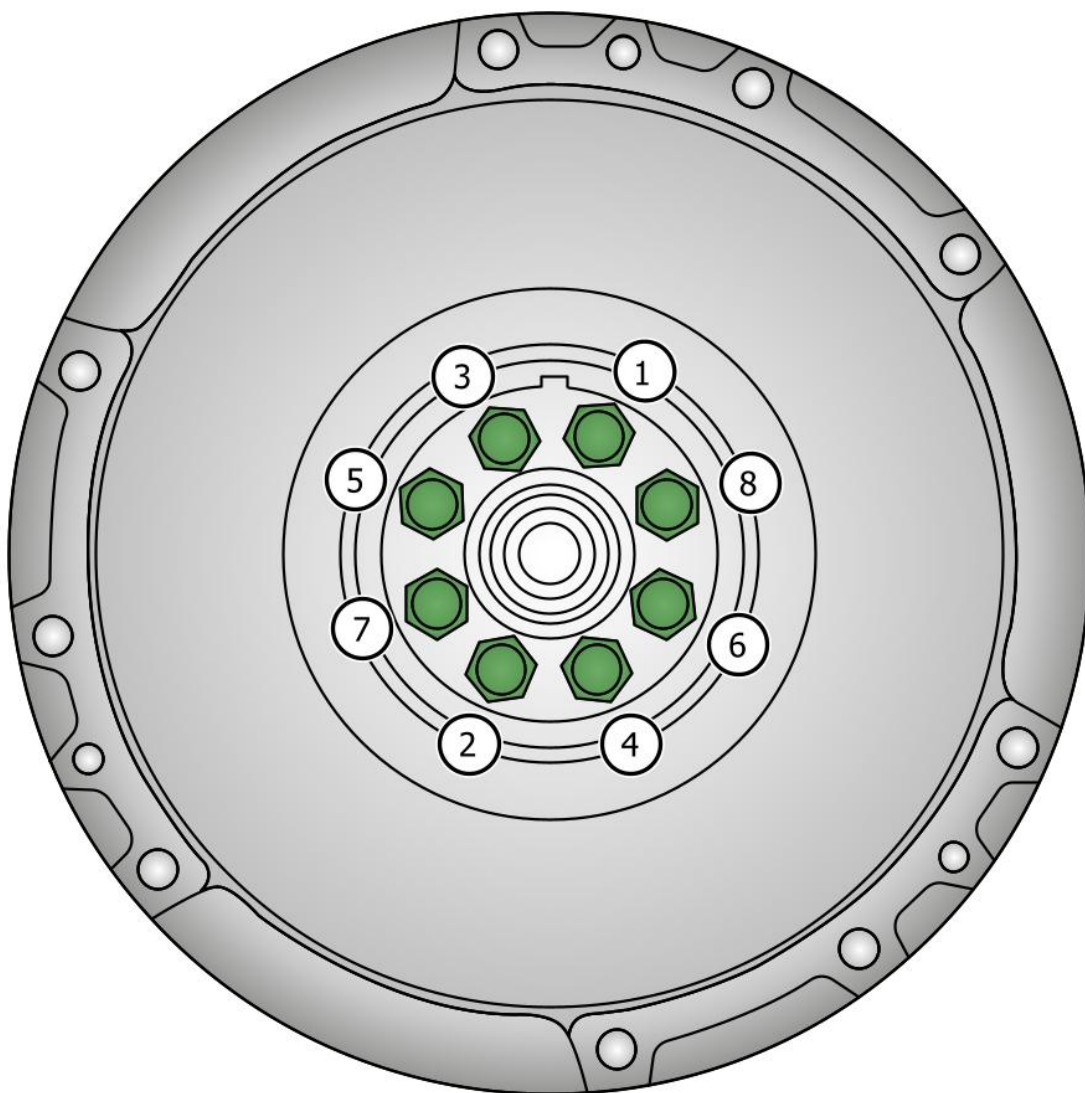
Stage 2 45 (°)

Drive plate

Drive plate

Renew the bolts

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Stage 1 45 (Nm)

Stage 2 45 (°)

Air-conditioning compressor

Air-conditioning compressor 38 (Nm)

Alternator

Alternator 38 (Nm)

Turbocharger

Turbocharger (To exhaust manifold) 19 (Nm)

Turbocharger (To crankcase) 22 (Nm)

Turbocharger oil return line

Turbocharger oil return line 8 (Nm)

Turbocharger oil supply line

Turbocharger oil supply line (Renew the seal)

Stage 1 10 (Nm)

Stage 2 45 (°)

Gearbox to engine:

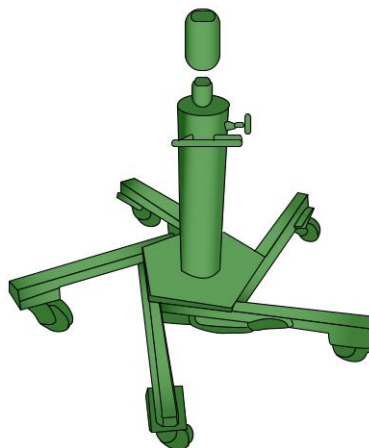
M12: (66 Nm)

M8: (19 Nm)

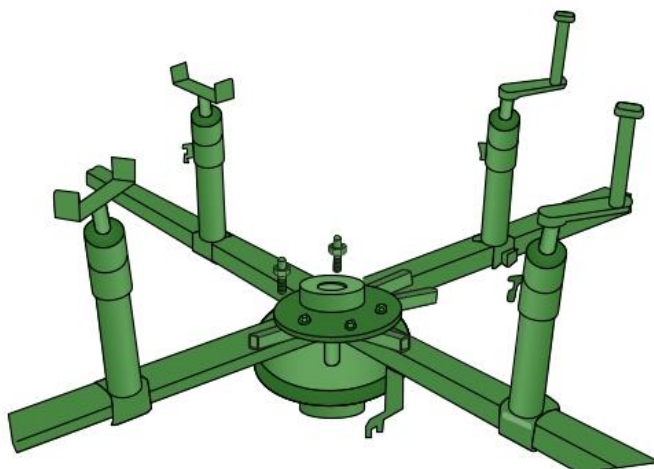
Special tools

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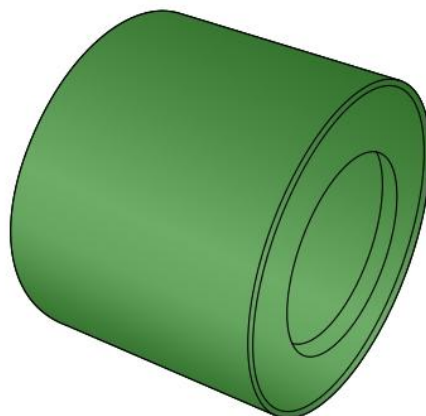
Transmission jack: (00 2 030)



Subframe support: (31 5 250)

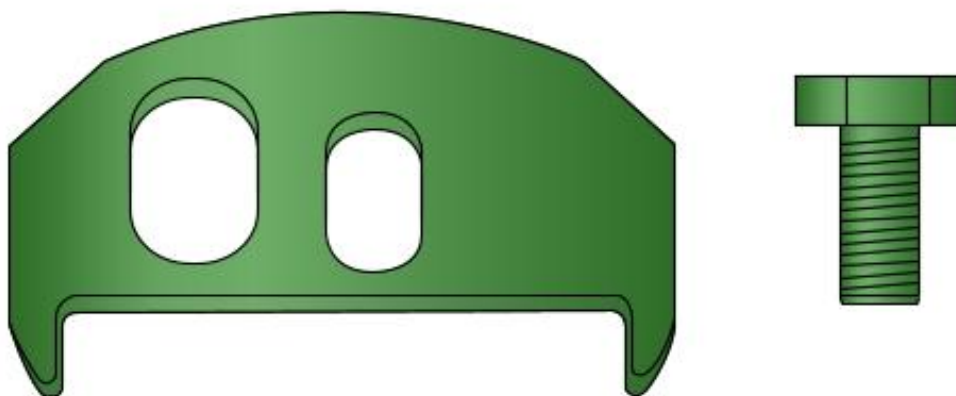


Bush: (11 9 265)

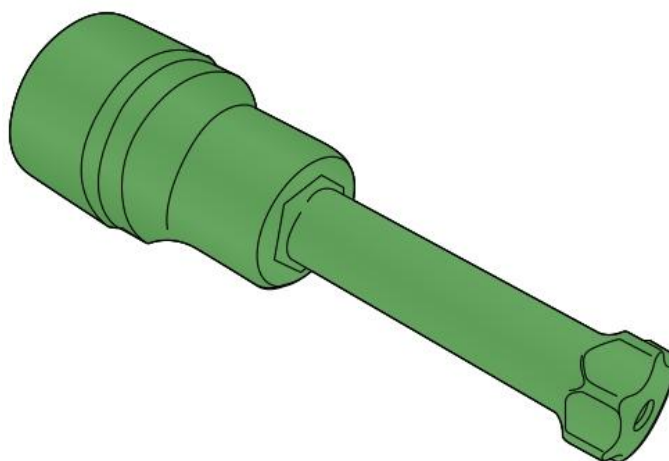


Flywheel locking tool: (11 9 811)

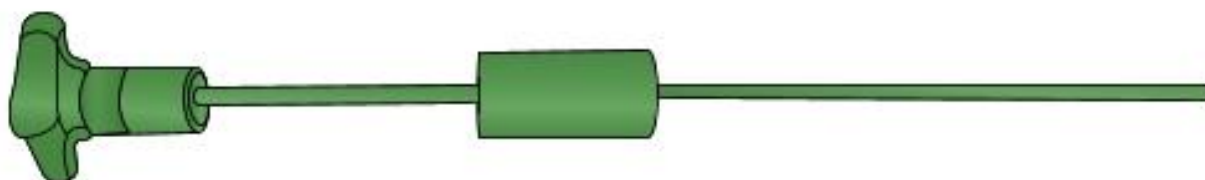
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Socket: (11 4 180)



Oil seal removal tool: (23 0 490)

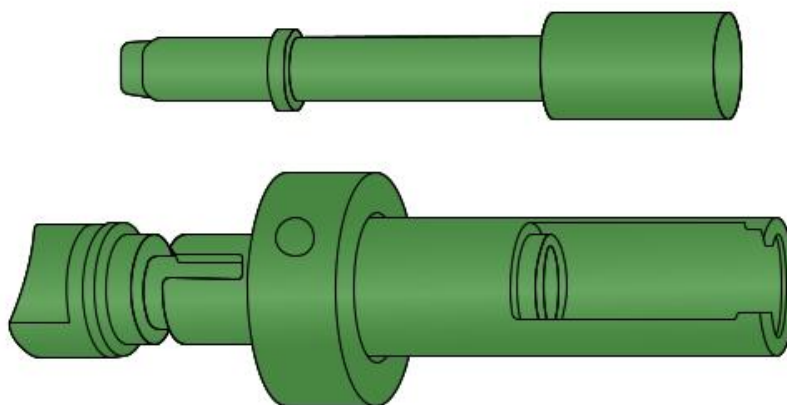


Subframe support: (2 184 136)

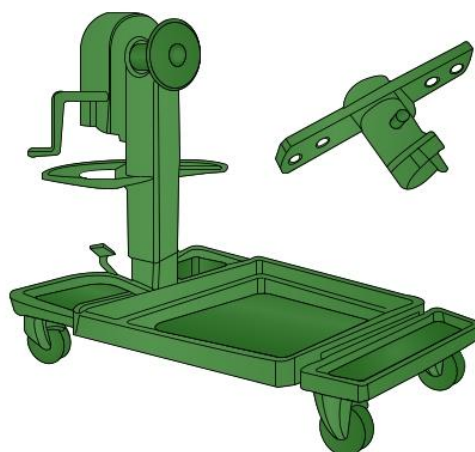
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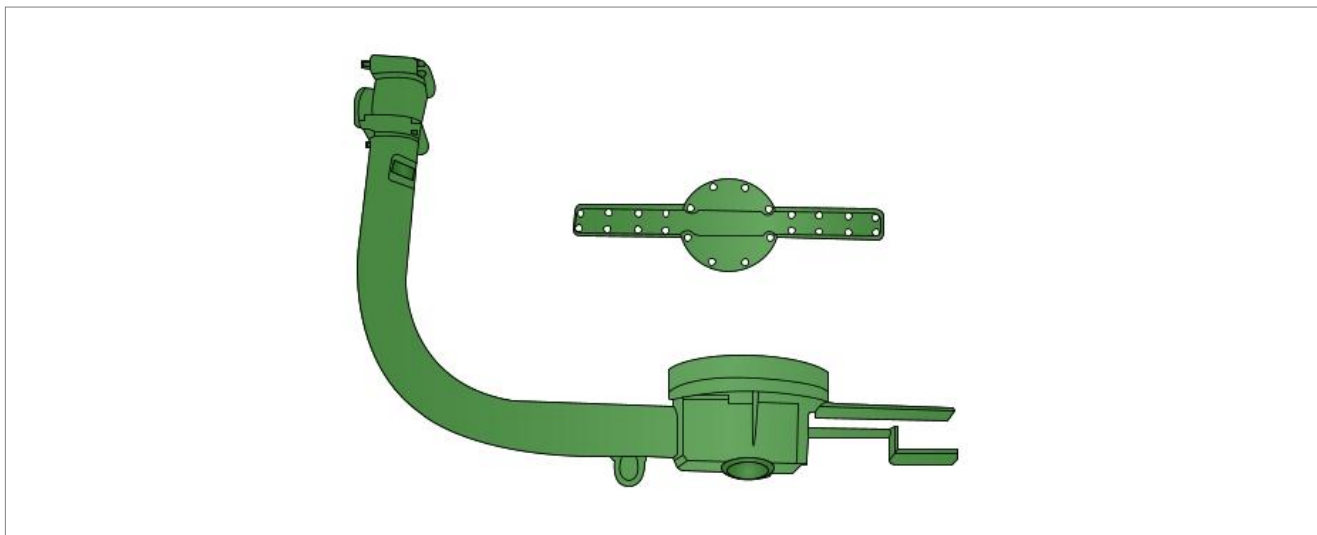
Blanking plug: (13 5 161, 13 5 162)



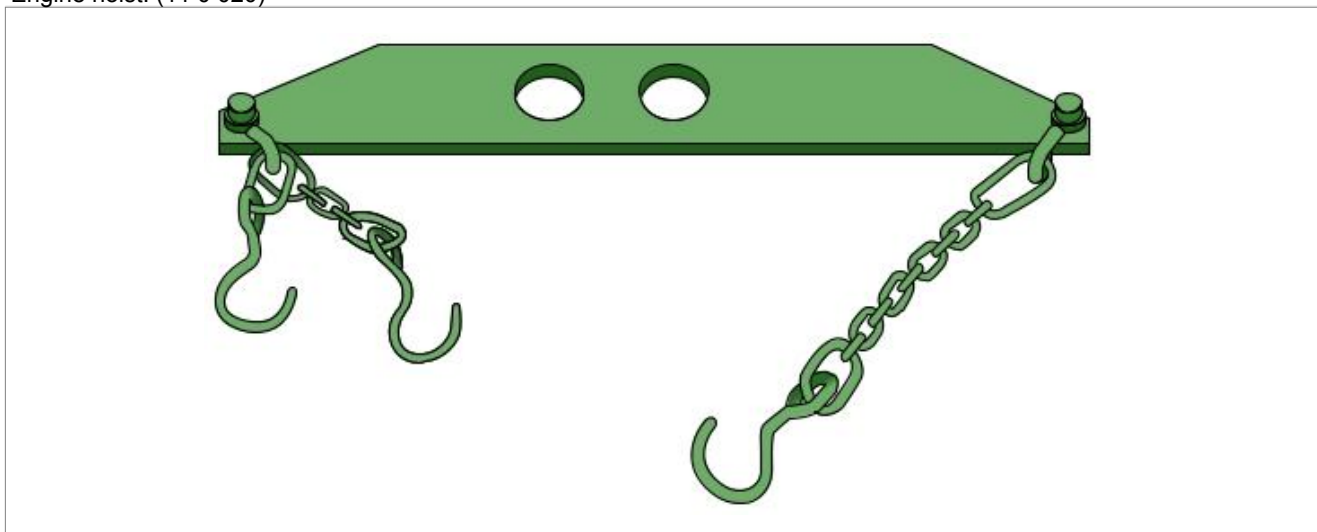
Engine support: (00 2 300)



Engine support: (00 2 320)

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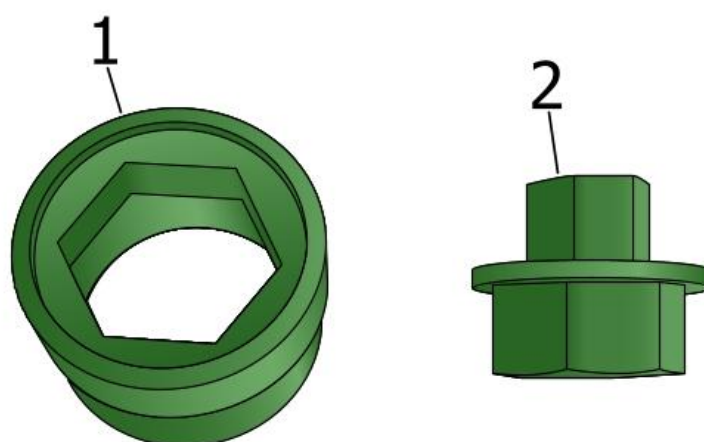
Engine hoist: (11 0 020)



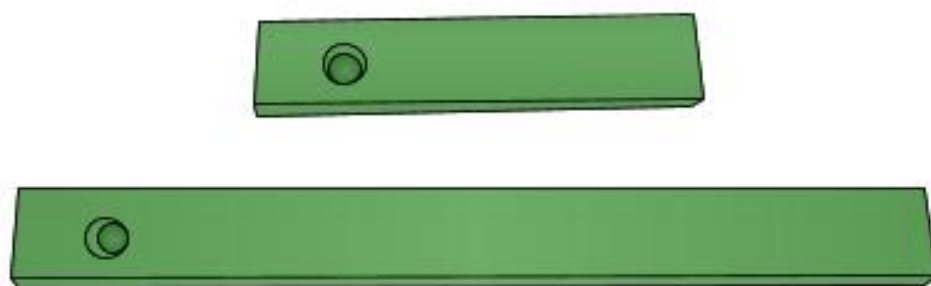
Locking pin: (11 4 120)



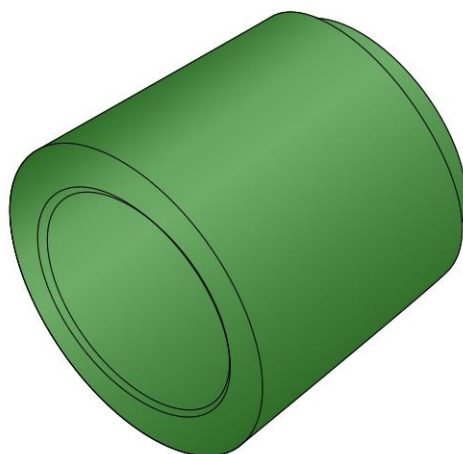
Extractor: (11 8 030)

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Special tools: (11 8 780)

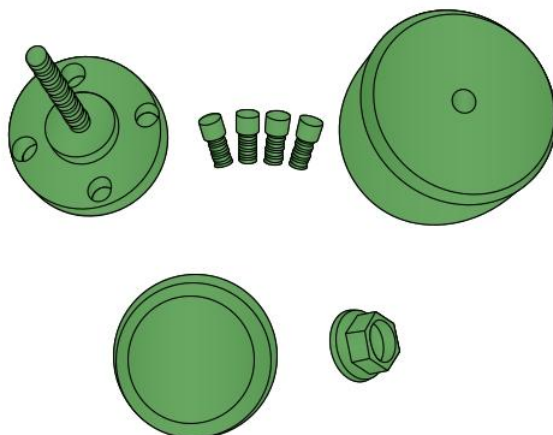


Oil seal fitting tool: (11 8 815)

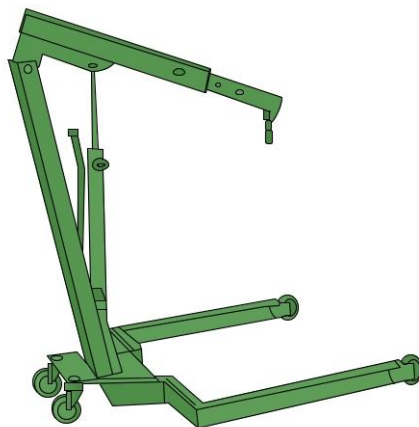


Oil seal fitting tool: (11 8 815, 11 8 814, 11 8 813, 11 8 812, 11 8 811)

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Engine lift: (2 220 718)

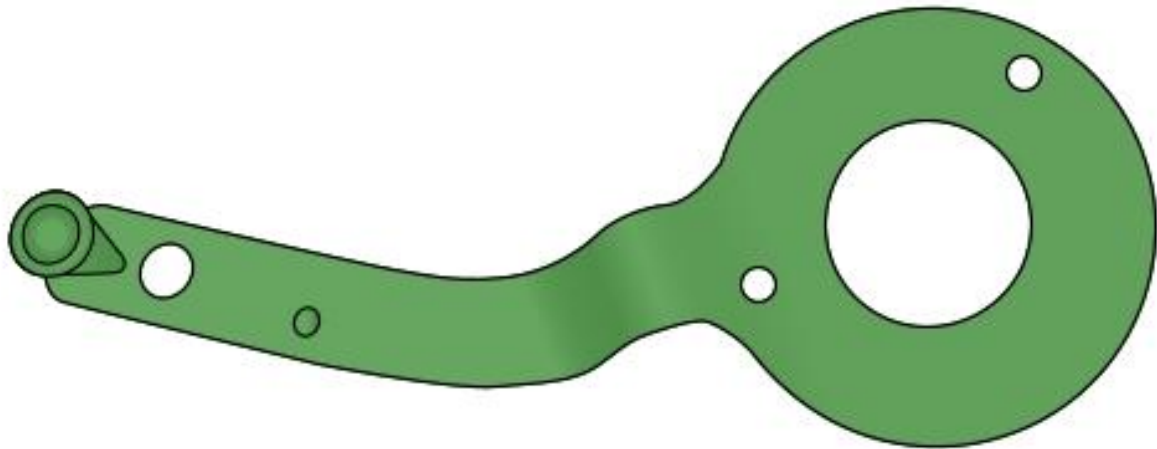


Flexplate locking tool: (2 288 380)

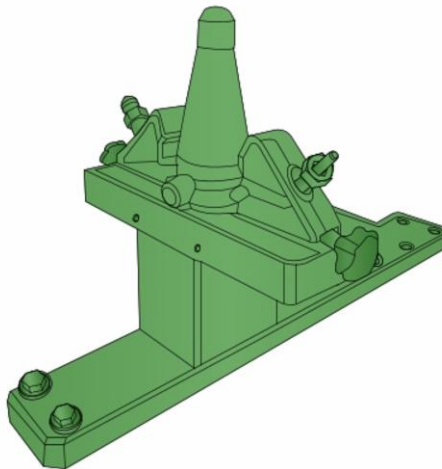


Crankshaft locking tool: (2 357 904)

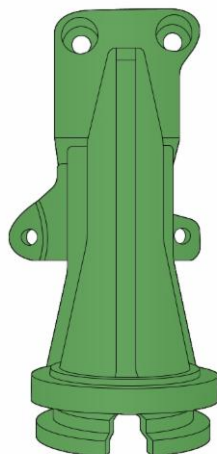
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Engine support: (2 413 082)

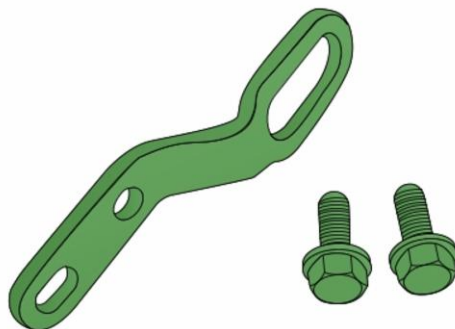


Engine support tool: (2 455 010)



Lifting eye: (2 459 012)

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Clutch compressing tool: (21 2 170)

